

A Second Part of Expansions of Sobolev Space Functions.

By Amlal El Mahrouss

September 23, 2026

1 Abstract

We present a framework useful for the expansion and exploitation of Sobolev Space operators in L-functions.

Keywords: PDE, Topology, Tensors, ML.

2 Definitions of an equation C^1

Definition 2.1. We suppose the Equation 1 of two natural norms f and g , in a Sobolev space for $k, p, i \in \mathbb{N}$ and $h, g \in \mathbb{R}$:

$$\left\|f\right\|_{k,p} = \sum_{i=0}^k \mathcal{N} = \sum_{i=0}^k \mathcal{N} + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (1)$$

Thus replacing $\sum_{i=0}^k \mathcal{N}$, with, $\left\|f\right\|_{k,p}$:

$$\left\|f\right\|_{k,p} + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (2)$$

Then replacing $\left\|f\right\|_{k,p}$ on the left with $\left(\left\|f\right\|_{k,p}\right)^n$

$$\left(\left\|f\right\|_{k,p}\right)^n + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (3)$$

Then assigning the Equation 3 to C^1 .

$$C^1 = \left(\left\|f\right\|_{k,p}\right)^n + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (4)$$

2.1 Complementary Remarks

Definition 2.2. Going back to Definition 2.1, we then construct the following for $k, p, i \in \mathbb{N}$ and $h, g \in \mathbb{R}$:

$$\left\|f\right\|_{k,p} = \sum_{i=0}^k \mathcal{N} = \sum_{i=0}^k \mathcal{N} + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (5)$$

And shifting $\left\|f\right\|_{k,p}$ to the right-hand-side:

$$\left\|f\right\|_{k,p} + \sum_{i=0}^k \mathcal{N} = \sum_{i=0}^k \mathcal{N} + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (6)$$

Squaring the summation term and taking its absolute value to ensure positivity, and moving it to the left-hand side:

$$\left\|f\right\|_{k,p} + \left|\sum_{i=0}^k \mathcal{N}^2\right| + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (7)$$

3 Highlighted Constants

Theorem 3.1. *We may introduce a constant term C^2 of:*

$$C^2 = \left\|f\right\|_{k,p} + \left|\sum_{i=0}^k \mathcal{N}^2\right| + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (8)$$

For $k, p, i \in \mathbb{N}$ and $h, g \in \mathbb{R}$.

Corollary 3.2. *We may introduce another equation per Theorem 8:*

$$C^{2+n} = (C^n) \times \left\|f\right\|_{k,p} + \left|\sum_{i=0}^k \mathcal{N}^2\right| + \frac{\left\|h\right\|_{k,p}}{\left\|g\right\|_{k,p}}. \quad (9)$$

For $k, p, i \in \mathbb{N}$ and $h, g \in \mathbb{R}$.